SAMUEL J. TUSICK

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PROFESSIONAL SUMMARY

Software engineering student (expected May 2027) working in an agile environment building artificial intelligence (AI) and automation tools used daily by an institute, co-designing a virtual patient training platform driven by a large language model (LLM) and tested by health professionals, and leading a team building a machine learning payload for a rocket flight. Led four student project teams. U.S. citizen, eligible to pursue a security clearance.

EDUCATION

Florida Gulf Coast University, Bachelor of Science in Software Engineering Expected completion May 2027

- In-major grade point average (GPA): 3.83 on a 4.0 scale. Overall GPA: 3.45 on a 4.0 scale. Dean's List: Fall 2023, Spring 2024, Fall 2024, and Spring 2025.
- Honors: Ruth M. and Cleveland L. Campbell Engineering Scholarship Endowed Fund, Kunkle Family Scholarship Endowed Fund.
- Relevant coursework: Machine Learning (in progress, Fall 2026), AI Agent Studio (in progress, Fall 2026), Software Architecture & Design (in progress, Fall 2026), Cloud Computing, Intro to Data Engineering, Data Structures & Algorithms, Software Engineering Fundamentals, Requirements Engineering & Analysis.

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, Java, JavaScript, TypeScript
- **Applied AI and Automation:** large language model (LLM) integration, prompt engineering, agentic workflow automation, retrieval-augmented generation (RAG), multi-service AI orchestration, neural networks and genetic algorithms
- **Cloud and Backend:** AWS (API Gateway, Lambda, Aurora, Cognito, Identity and Access Management, Cloud Development Kit, Transcribe), Docker, Flask, REST application programming interfaces (APIs), PostgreSQL (pgvector), Supabase, SQLite
- **Frontend:** React, Vite, Tailwind CSS
- **Tools:** Git, GitHub Actions, Selenium, TestNG, Jira, Confluence, Azure DevOps, Power Automate, NumPy, Pygame, Gymnasium, OpenRocket, Agile methodology

EXPERIENCE

Student Research Assistant March 2025 to Present
Florida Gulf Coast University, Dendritic AI and Data Science Institute (on-site) **20 hours per week**

- Cut course module production time about 50% (about 90 to 45 minutes) for the university's AI Virtual Academy (AIVA) by building two large language model (LLM) tools: one turns faculty PDFs and slide decks into a reviewed narration script, and the other turns the approved script into an avatar video and slide deck. A cost estimate and approval step runs before any credits are spent. Used by the development and infrastructure teams.
- Created the institute's first workflow handbook (25 pages, used by all 10 team members) with a tool that writes and updates Confluence pages in one fixed template, checks for duplicates, screens out private data, then validates each page and reads it back to confirm it published correctly. Every page puts the owner, steps, tools, and review rules in the same place.
- Gave the whole team one standup-style view every morning by automating a daily Jira summary posted as a Microsoft Teams card at 9 AM, covering every task on the board, running since October 2026.
- Replaced texting between teams, where updates were often lost, with Power Automate alerts that post to Teams when files land in about 25 monitored folders, for the Digital Media and Design and Infrastructure teams.
- Built an AI tool that turns a topic into a finished 1080p explainer video with voiceover in about 25 minutes, replacing a day or two of Digital Media and Design team time, using the team's exact content and branding guidelines.
- Helped plan and design the 2026 Summer AI Academy (500+ participants, 150+ instructional hours), ran day-to-day operations, and taught a GitHub Actions lesson in the Software Testing course.
- Built 14 custom AI tutors covering 13 engineering courses for the Engineering Learning Hub, used by students seeking tutoring help and by Learning Hub tutors.

Subteam Lead and Backend Developer, Router-Level Security Detection (lab class group project) February 2025 to April 2025
Florida Gulf Coast University, Eagle CyberNest Lab (on-site) **5 hours per week**

- Led backend architecture decisions and task coordination for a 5-member team, delivering the project by its April 29

deadline.
- Built a Flask web application that gathers CPU, memory, bandwidth, and connected-device data from 3 router models over Secure Shell using 7 analysis commands, and serves it through a REST application programming interface (API) to a dashboard. Stored results in SQLite, and tested and deployed the router-communication scripts.

PROJECTS

MAIVA: AI Virtual Patient Training Platform | TypeScript, JavaScript, React, AWS, LLM API August 2026 to Present

- Co-designed, with a graduate researcher, a platform where students practice clinical interviews with artificial intelligence (AI) simulated patients, built from requirements set by two clinical assistant professors. Chat and voice modes are working and have been tested by 3 professionals, including Marieb's dean. Instructors build their own patient scenarios and every transcript is saved for review.
- Co-designed the orchestrator that connects the user interface, speech-to-text, text-to-speech, the large language model (LLM) application programming interface (API), and Amazon Web Services (AWS) services, streaming each patient reply sentence by sentence, saving every turn in real time, and rebuilding a session from its saved transcript after a restart.
- Built the React frontend from scratch, integrated AWS Transcribe for speech input, and wrote the infrastructure endpoints and Cloud Development Kit (CDK) in TypeScript plus the Structured Query Language (SQL) data layer.
- Co-built Local, Development, and Test environments on a serverless AWS backend (API Gateway, Lambda, Aurora, Cognito, Identity and Access Management, CDK).
- Avatar mode is in development: 3D patient models are built in Unreal Engine 5, and the team is designing how to drive their facial animation from text-to-speech viseme data.

Real-Time Landing Site Detection with Machine Learning | Python, OpenRocket October 2026 to Present

- Team lead for a 4-person team selected as one of two winning teams in the Florida Gulf Coast University Payload Challenge, earning $500 in funding and a flight on the club rocket, earliest January 2027.
- Ran a proof of concept with a pretrained segmentation model that proposed 3 landing zones with a safety percentage for every frame; now planning to train our own model on aerial datasets and run it on a Raspberry Pi with a 4K camera.
- Chose to test during powered ascent rather than parachute descent, to stress the model with flame, vibration, spin, and motion blur. Set targets of at least 2 predictions per second and 80% pixel accuracy, to be reported whether met or not.
- Built an OpenRocket flight simulation of the club rocket.

Email Summary Agent (personal project) | Python, AWS Lambda, Docker, PostgreSQL (pgvector), Microsoft Graph API August 2026 to Present

- Built and deployed an agent that runs daily across 2 Outlook accounts, summarizing every email from the last 24 hours with an urgency level using retrieval over embedded email content. Runs on a schedule in a Docker container on AWS Lambda, with failure alerts and secrets in AWS Secrets Manager, and data in Supabase.

Machine Learning Self-Driving Car Simulator | Python, Pygame, Gymnasium

- Building a neural network and genetic algorithm trainer from scratch in Python, with no machine learning libraries, to teach a car agent to complete a track; now tuning mutation rate, sigma, and pool fraction.
- Built the simulation: steering-based car physics, pixel-mask collision detection, five ray-cast distance sensors, a checkpoint system, and a Gymnasium environment.

Switch4Good Database System | PostgreSQL, SQL January 2026 to May 2026

- Led a 3-person class team that designed a 19-table PostgreSQL database with 3 access roles for the nonprofit Switch4Good, sized for its 100+ universities and 1,500+ students and now used by the client.
- Led client communication, assigned tasks, ran reviews, and wrote most of the documentation, schema, and design work.

LEADERSHIP AND INVOLVEMENT

Deeptech Club, Member since June 2026, Payload Team Lead since October 2026 June 2026 to Present
Computer Science & Software Engineering Club, Member August 2024 to Present
Florida Gulf Coast University AI Day, Panelist, "AI in University Operations" March 2026
Led four student project teams: Eagle CyberNest, Switch4Good, Pinterest Testing, Landing Site Payload 2025 to Present
Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate (valid until August 29, 2027) August 2025
Volunteer: campus food pantry (12.5 total hours), United Way Youth Oasis (5 total hours), Wonder Gardens (5 total hours) Fall 2024 to Fall 2026